

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT **Paper -2(Adv-2020-P2-Model)** **Date: 04-01-2026**

Time: 02.00Pm to 05.00Pm **GTA-10** **Max. Marks: 198**

04-01-2026_Sr. XL-500, Sr.S60(N&S), Elite, IAS & LIIT_Jee-Adv(2020-P2)_GTA-10_Syllabus,

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

MATHEMATICS : TOTAL SYLLABUS

Name of the Student: _____ **H.T. NO:** | | | | | | | | |

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JEE-ADVANCED-2020-P2-Model

Time: 3:00Hour's **IMPORTANT INSTRUCTIONS** **Max Marks: 198**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 6)	Questions With Integer Answer Type	3	-1	6	18
Sec – II(Q.N : 7 – 12)	**Questions with Multiple Correct Choice +1 partial marks**	4	-2	6	24
Sec – III(Q.N : 13 – 18)	Questions with Numerical Value Answer Type	4	0	6	24
Total				**18**	**66**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 24)	Questions With Integer Answer Type	3	-1	6	18
Sec – II(Q.N : 25 – 30)	**Questions with Multiple Correct Choice +1 partial marks**	4	-2	6	24
Sec – III(Q.N : 31 – 36)	Questions with Numerical Value Answer Type	4	0	6	24
Total				**18**	**66**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 37 – 42)	Questions With Integer Answer Type	3	-1	6	18
Sec – II(Q.N : 43 – 48)	**Questions with Multiple Correct Choice +1 partial marks**	4	-2	6	24
Sec – III(Q.N : 49 – 54)	Questions with Numerical Value Answer Type	4	0	6	24
Total				**18**	**66**

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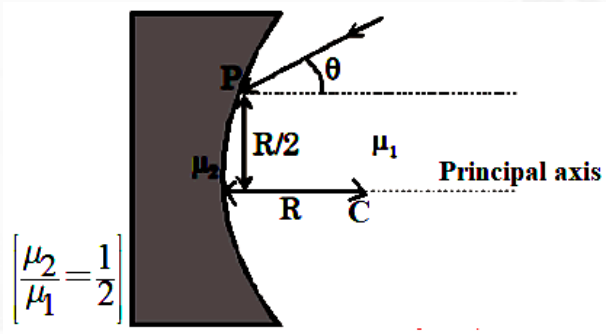
PHYSICS **Max Marks: 66**

SECTION-I
(INTEGER ANSWER TYPE)

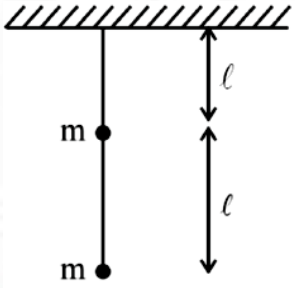
- This section contains SIX (06) questions.
- The answer to each question is a SINGLE DIGIT INTEGER ranging from 0 TO 9, BOTH INCLUSIVE.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual Numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated **according to the following marking scheme:**

Full Marks : +3 If ONLY the correct integer is entered;
Zero Marks : 0 If the question is unanswered;
Negative Marks : -1 In all other cases

1. If the minimum value of θ so that the ray gets totally reflected from spherical surface of radius of curvature of R is 5n radians, find n.



2. Two particles of m each are fixed to a massless rod of length 2ℓ. The rod is smoothly hinged at one end to ceiling. It performs oscillations of small angle in vertical plane. If the length of the equivalent simple pendulum is $\frac{5\ell}{x}$, find the value of x.



3. A sample of hydrogen atoms are in their ground state and stationary, when a beam of neutrons of kinetic energy 25.5 eV collides with them. Take mass of hydrogen atom = mass of neutron = 1.6×10^{-27} kg, $1eV=1.6\times10^{-19}J$. If the collision is head–on, find the maximum possible number of photons of different wavelengths that can be emitted after the collision, from the hydrogen atoms. Assume that only a single collision is possible with a hydrogen atom.

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4. Two particles are at same position at t = 0. The first particle moves with velocity $v=\left(3t^2+8t\right)m/s$ and the second one moves with velocity $v=(10t+20)m/s$. At what instant (in s) will they meet?

5. One mole of a monoatomic ideal gas is subjected to a thermodynamic cycle shown in the Temperature–Entropy (T–S) diagram. The cycle consists of three processes connecting states A, B and C.

The states are defined by their entropy and temperature coordinates as:

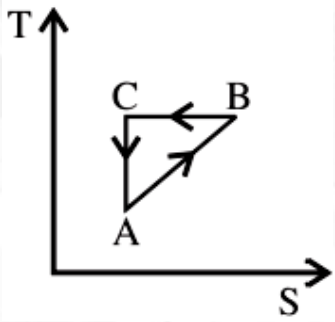
States A: (S_0,T_0)

States B: $(2S_0,2T_0)$

States C: $(S_0,2T_0)$

The processes are as following

1. $A\rightarrow B$: A process where temperature increases linearly with entropy.
2. $B\rightarrow C$: An isothermal compression process.
3. $C\rightarrow A$: An adiabatic (isoentropic) expansion process.



For this cyclic process, the ratio of the magnitudes of the heat rejected during the isothermal process to the net work done by the gas is found to be an integer Z. Find the value of Z.

6. The circular scale of a screw gauge consists of 50 divisions and the linear distance travelled by screw in one complete rotation is equal to 0.5 mm and it is equal to the value of one pitch scale division. When nothing is placed between the studs, 46th division of circular scale coincides with the index line of pitch scale and zero of the pitch scale is just inside the circular scale. Now a thin plate of thickness 4.78 mm is inserted between the studs by unscrewing the circular scale and fit it just right to measure its thickness. The minimum number of rotations to be done by the ratchet in order to insert the plate during unscrewing is 2n. Find n.

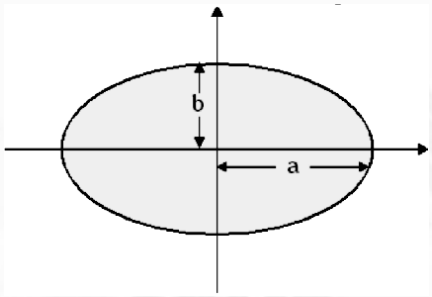
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SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

•This section contains SIX (06) questions.
•Each question has FOUR options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
•For each question, choose the option(s) corresponding to (all) the correct answer(s).
•Answer to each question will be evaluated **according to the following marking scheme:**
Full Marks: +4 If only (all) the correct option(s) is(are) chosen; *Partial Marks* +3 If all the four options are correct but ONLY three options are chosen;
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
Partial Marks: +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: -2 In all other cases.

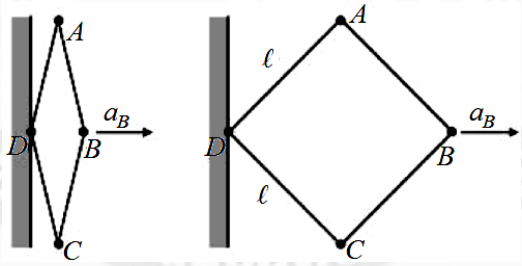
7. Moment of inertia of an elliptical disc of semi–axes a and b, having mass m about



A) the major axis is $\frac{mb^2}{4}$ **B)** the major axis is $\frac{mb^2}{2}$

C) the minor axis is $\frac{ma^2}{2}$ **D)** the minor axis is $\frac{ma^2}{4}$

8. The rhombus ABCD consists of four rigid rods each of length ℓ connected by joints of negligible size, which allow the angles between the rods to change, while the rods remain in one plane. The rhombus is fixed at point D and at the beginning, the point B is located very close to point D. Let point B now move with constant acceleration of magnitude a_B to the right. At the instant when the rhombus takes the shape of square,

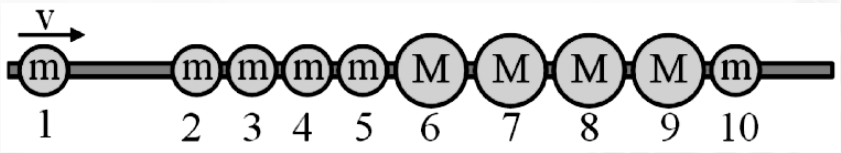


A) the speed of B is $2^{\frac{3}{4}}\sqrt{a_B\ell}$ **B)** the speed of A is $2^{\frac{1}{4}}\sqrt{a_B\ell}$

C) the acceleration of A is $\left(\frac{13}{2}\right)^{\frac{1}{3}} a_B$ **D)** the acceleration of A is $\left(\frac{13}{2}\right)^{\frac{1}{2}} a_B$

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9. Four balls each of mass m and another four balls each of mass M and then one more ball of mass m are strung on a horizontal, frictionless, fixed rod, as shown in the figure. The balls are close to each other, they are made of perfectly elastic material and m < M. From the left, another ball of mass m is coming at a speed of v and collides with the first one of the balls string. Which of the following statements are correct?



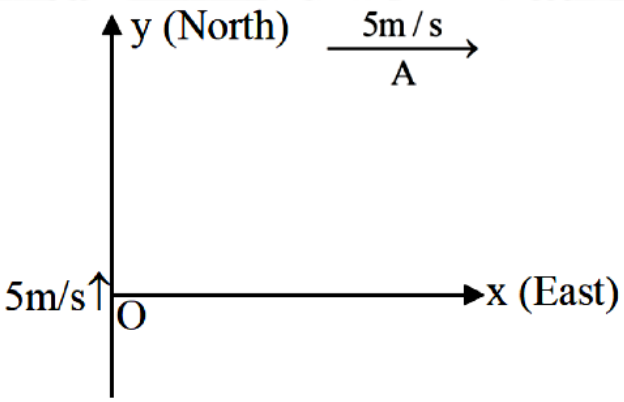
A) The final velocity of ball 1 is zero

B) The final velocity of ball 5 is zero

C) The final velocity of ball 9 is $\frac{m(M-m)v}{(M+m)^2}$

D) The final velocity of ball 10 is $\frac{4Mmv}{(M+m)^2}$

10. A cross country runner at point O is running at 5 m/s towards North. Coefficient of friction between his shoes and ground is 0.1. He changes his direction of running in time T so that now he is at point A, running at 5 m/s towards East. $\left(g=10ms^{-2}\right)$. Which of the following statements are correct?



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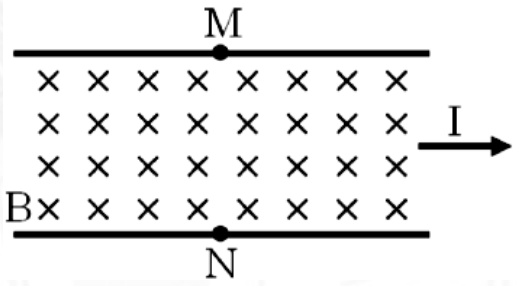
A) To keep T minimum, he must keep running at constant speed between O and A.

B) Minimum value of T rounded off to nearest integer is 7 seconds.

C) Minimum value of T cannot be found as coordinate of point A are not specified.

D) If T is kept minimum, equation of path of runner is $y = 5\left(2^{\frac{3}{4}}\right)\sqrt{x} - x$.

11. In the well–known Hall experiment, a thin conducting strip carries current as shown in the figure. A uniform magnetic field B is also applied perpendicular to the plane of the strip and directed into the page. Inside the strip, due to the electric current, positive charges may be moving to the right. This may also be described equivalently as the motion of negative charges to the left. As a result, a potential difference can arise between two points on the strip M and N across its width. Let $\Delta V = V_M - V_N$. Based on this, which of the following statements are incorrect?



A) $\Delta V > 0$ if the electric current is due to the motion of negative charges.

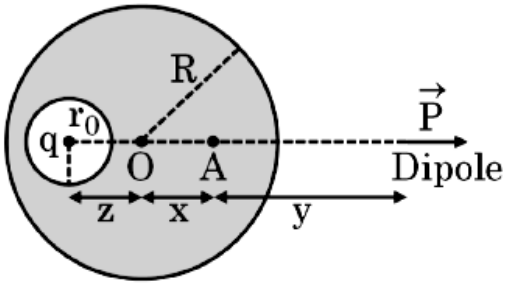
B) $\Delta V > 0$ if the electric current is due to the motion of positive charges.

C) $\Delta V > 0$ under all conditions.

D) $\Delta V < 0$ under all conditions.

12. A solid conducting sphere of radius R has a spherical cavity of radius r_0. A point charge q is placed at center of the cavity. A dipole having dipole moment $\vec{P}$ is placed outside the sphere as shown in the figure.

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Choose INCORRECT statements.

A) Force on q due to the dipole is zero.

B) Force on q due to the dipole is $\frac{Pq}{2\pi\varepsilon_0(x+y+z)^3}$.

C) Potential at point A due to charges induced on outer surface of the sphere is $\frac{q}{4\pi\varepsilon_0 R}$.

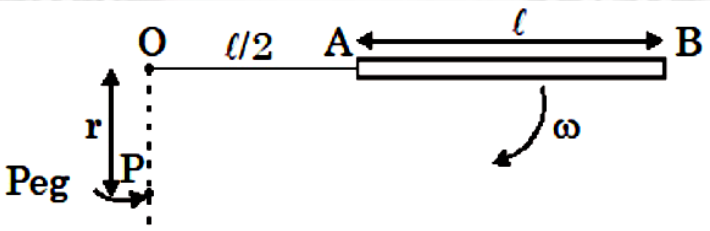
D) Potential at point A is $\frac{q}{4\pi\varepsilon_0 R}-\frac{P}{4\pi\varepsilon_0 y^2}$.

SECTION – III
(NUMERICAL VALUE TYPE)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual Numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4 If ONLY the correct numerical value is entered;
Zero Marks: 0 **In** all other cases..

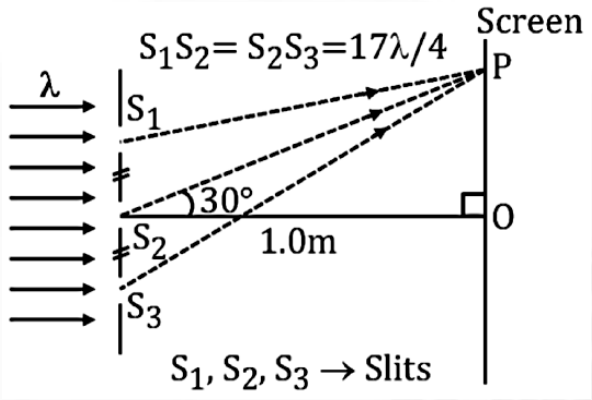
13. A thin uniform rod AB of mass M and length ℓ attached to a string OA of length $\ell/2$ is placed on a smooth horizontal plane and rotates with angular velocity ω around a vertical axis through O. A peg P is inserted into the plane in order that, on striking it, the rod will come exactly to rest.



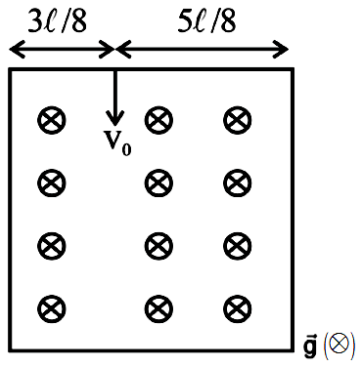
Location of peg for rod coming to rest is r = _____ m (ℓ = 12 m)

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14. Intensity reaching P from each source (narrow slits S_1, S_2, S_3) is I. The resultant intensity at P is $\alpha I(\lambda <<< 1m)$. The value of α is _____

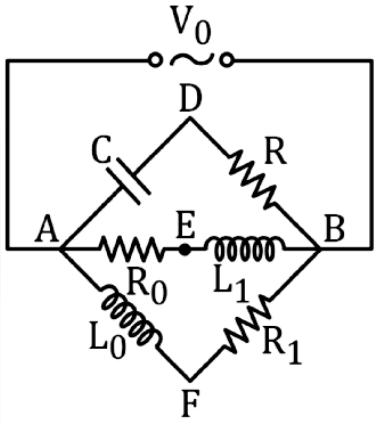


15. A particle of mass m and charge q is currently located on the interior of a square room of side length ℓ as shown in the figure below (top view). A uniform magnetic field B is directed into the page in the figure. Suppose that particle is now projected at an initial velocity $V_0 = \frac{qB\ell}{8m}$ perpendicular to a wall of the room, as depicted in the figure. If the time it requires to return to its initial position is $t = \frac{k\pi m}{qB}$, find the value of k, assuming that its collisions with the walls are perfectly elastic.

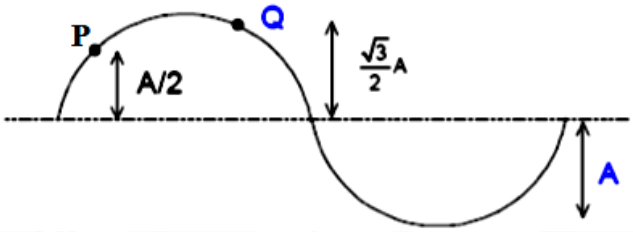


16. Consider the following AC circuit. The voltage difference between D and E has amplitude $V_{DE} = 7V$. Similarly $V_{DF} = 15V$ and $V_{EF} = 20V$. What is magnitude of V_0 (in V)? (V_0 is amplitude of source voltage)

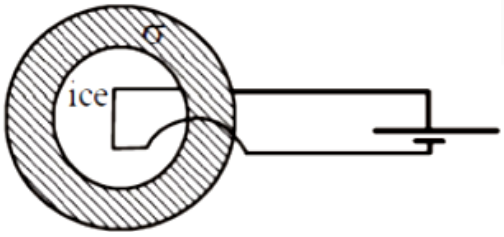
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17. Figure shows the snapshot of the transverse sinusoidal wave of amplitude A on a string at time t = 0. Frequency of wave is $\frac{125}{6}Hz$. T_1 is the time at which the line joining P and Q become horizontal for the first time (after t = 0) if wave is travelling towards +ve x direction and T_2 is the time at which the line P and Q becomes horizontal for first time (after t = 0) if wave is travelling in –ve x direction. Then $\frac{T_1}{T_2} =$ _____



18. Figure shows a thick spherical shell made of electrical conductivity σ and has inner and outer radii of 10 cm and 20 cm respectively. It is filled with ice inside it. Its inside and outside surfaces are kept at different potentials by a battery of internal resistance $\frac{2}{\pi}\Omega$ and emf $\varepsilon = 5V$. Find the value of σ (in SI units) for which ice melts at maximum possible rate if 25% of heat generated by shell due to joule heating is used to melt the ice.



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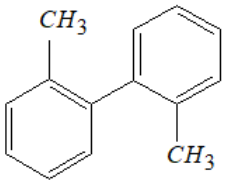
CHEMISTRY **Max Marks: 66**

SECTION-I
(INTEGER ANSWER TYPE)

- This section contains SIX (06) questions.
- The answer to each question is a SINGLE DIGIT INTEGER ranging from 0 TO 9, BOTH INCLUSIVE.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual Numeric keypad in the place designated to enter the answer.
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Full Marks: +3 If ONLY the correct integer is entered;
Zero Marks: 0 If the question is unanswered;
Negative Marks: -1 In all other cases

19. Degree of unsaturation of compound [R] of the following reaction sequence is 2X. What is the value of X?



$$\xrightarrow[2.KCN/DMSO(2\,equiv.)]{1.NBS/CCl_4(2\,equiv.)} [Q] \xrightarrow[2.anhydr.AlCl_3(2\,equiv)]{1.SOCl_2(2\,equiv.)} [R]$$

20. Consider the following partially balanced redox reactions:

(I) Hydrolysis of XeF_4 (coefficients partly given):

$$XeF_4 + H_2O \rightarrow 2XeO_3 + 4Xe + O_2 + HF$$

(II) Reaction of F_2 with $NaOH$ (coefficients partly given):

$$F_2 + NaOH \rightarrow OF_2 + NaF + 1H_2O\,.$$

In Experiment 1, 0.75 mol of XeF_4 reacts according to reaction (I).

In Experiment 2, y mol of F_2 reacts according to reaction (II).

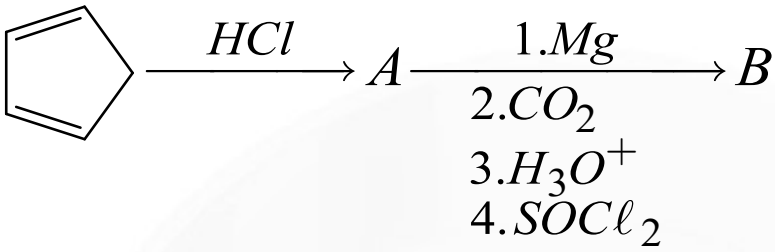
If the total moles of electrons lost are equal in both experiments, what is the value of y?

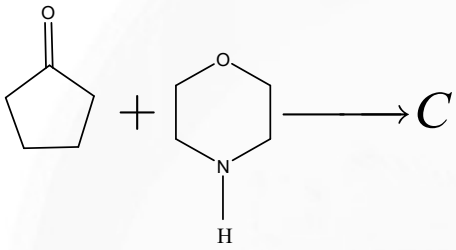
21. 2 moles of an ideal gas with the adiabatic exponent 5/3 goes through a process, $PV^2 =$ constant, as a result of which the absolute temperature of the gas increases 2–fold. The entropy increment of the gas in this process is **Z** times R ln2. What is the value of Z?

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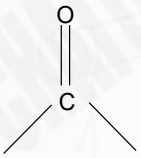
22.





$$B + C \longrightarrow I \xrightarrow{hydrolysis} D[Major] + \begin{bmatrix} Nitrogen\,containing \\ compound \end{bmatrix}$$

Number of '$-OH$' in D = y.
Number of chiral carbons in D = z.
Double bond equivalent of 'D' = P.



Number of groups in D = X
Find $\left(x[y+z]+P\right)$?

23. A gaseous mixture of H_2 and O_2 in 2: 1 mole ratio is used to prepare water vapour by the following reaction.

$$2H_2(g) + O_2(g) \rightarrow 2H_2O(g)$$

The total pressure of gases in the container is 4.5 atm at $57^\circ C$ before the reaction. The final total pressure of gases (in atm) at $127^\circ C$ after reaction assuming 80% yield of water vapour is_____

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24. How many of the following can liberate dihydrogen gas when reacted with B_2H_6?

i) CH_3OH ii) $HC\ell / AlCl_3$ iii) $aq.KOH$

iv) CO v) $P(CH_3)_3$ vi) excess NH_3 / Δ

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

•This section contains SIX (06) questions.
•Each question has FOUR options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
•For each question, choose the option(s) corresponding to (all) the correct answer(s).
•Answer to each question will be evaluated **according to the following marking scheme:**
Full Marks: +4 If only (all) the correct option(s) is(are) chosen; *Partial Marks* +3 If all the four options are correct but ONLY three options are chosen;
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
Partial Marks: +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: -2 In all other cases.

25. A water sample suspected to contain nitrite ion $\left(NO_2^-\right)$ is treated with sulphanilic acid and N-(1-naphthyl)ethylenediamine in cold, mildly acidic medium, forming a pink azo dye.

Which statements are correct?

A) NO_2^- forms HNO_2, which diazotises sulphanilic acid to generate an aromatic diazonium ion.

B) The diazonium ion must be produced fresh and dilute, since concentrated diazonium solutions are unsafe and unstable.

C) Azo coupling occurs by electrophilic aromatic substitution on the activated naphthylamine ring.

D) Degree of unsaturation of dye of pink colour is 10

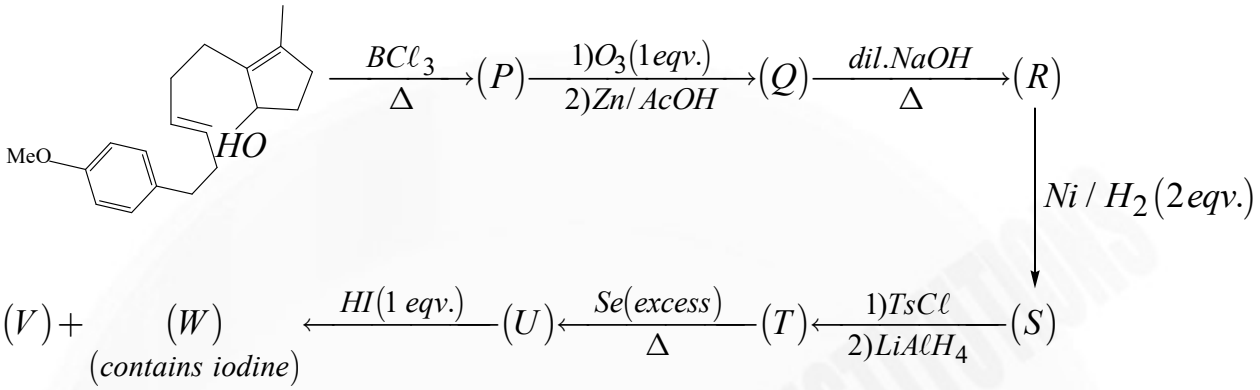
26. Which of the following can oxidize $KI(aq)$?

A) $Cr_2O_7^{2-}$ **B)** MnO_4^- **C)** $FeC\ell_3$ **D)** $CuSO_4$

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27. Consider the following reaction sequence:



[Me = Methyl group; Ac = Acetyl group and Ts is Tosyl group and assume that 'P' to 'W' are the major organic products]

Now, select the correct statement(s) regarding above sequence of reactions:

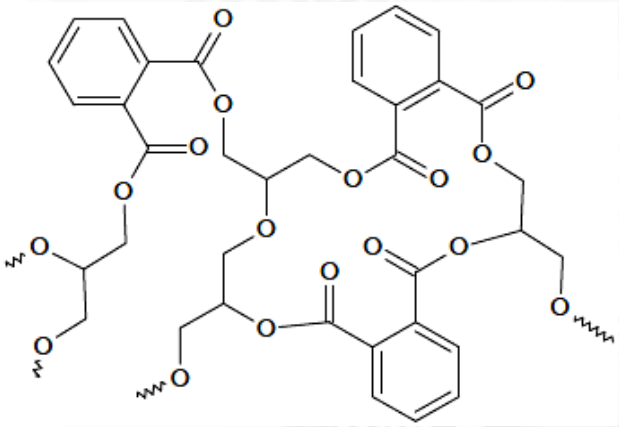
A) Sum of degree of unsaturation of (**T**), (**U**) and (**V**) is 33.

B) (**Q**) on reaction with Mg (Hg) in ether followed by hydrolysis gives a compound which on treatment with methanal in presence of dry HCl gas, gives another product with degree of unsaturation 8.

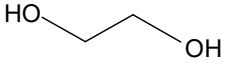
C) (**R**) contains 3 chiral carbon atoms.

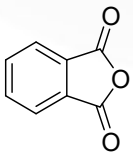
D) Both (**R**) and (**S**) gives positive Lucas test.

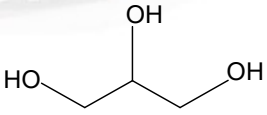
28.

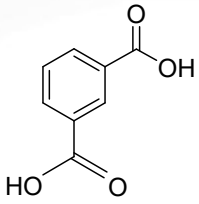


Find the correct monomer for the above given polymer.

A) 

B) 

C) 

D) 

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29. The correct statement(s) among the following is/are:

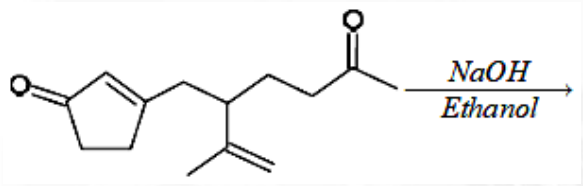
A) Ionization energy of N_2 molecule is more than that of O_2 molecule.

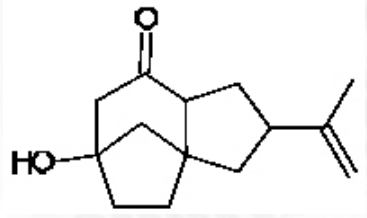
B) From N_2 molecule the energy required to remove an electron from σ_{2P_z} is less than to remove an electron from π_{2P_y} or π_{2P_x}.

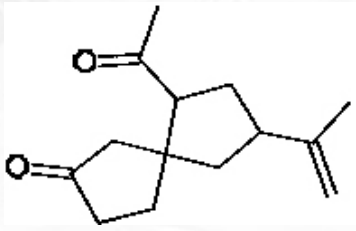
C) O-O bond is weaker than S-S bond but O=O bond is stronger than S=S.

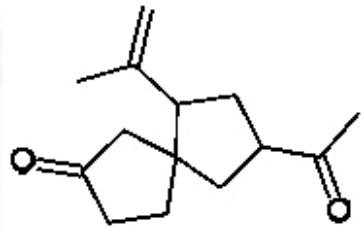
D) Bond energy of N_2^+ is more than that of N_2^-

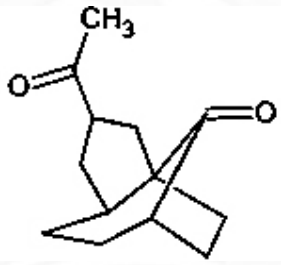
30. Which of the following is/are formed in the sequence of reaction?



A)


B)


C)


D)


SECTION – III
(NUMERICAL VALUE TYPE)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual Numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4 If ONLY the correct numerical value is entered;
Zero Marks: 0 **In** all other cases..

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31. In a bio-cell, the cathode contains an aqueous solution of $K_3\left[Fe(CN)_6\right]$. Ferricyanide acts as the cathodic electron acceptor according to: $\left[Fe(CN)_6\right]^{3-} + e^- \rightleftharpoons \left[Fe(CN)_6\right]^{4-}$

Initially, the concentrations of $\left[Fe(CN)_6\right]^{3-}$ and $\left[Fe(CN)_6\right]^{4-}$ are equal.

Due to an accidental contamination, Fe^{2+} ions leak into the cathode chamber. They react with ferricyanide in a qualitative analysis type reaction, leading to the formation of an insoluble deep blue complex (Prussian blue), effectively removing $\left[Fe(CN)_6\right]^{3-}$ from the solution.

Assume that after contamination, the concentration of $\left[Fe(CN)_6\right]^{3-}$ becomes one-tenth of its initial value, while the concentration of $\left[Fe(CN)_6\right]^{4-}$ remains practically unchanged.

Given:

Standard reduction potential for $\left[Fe(CN)_6\right]^{3-} / \left[Fe(CN)_6\right]^{4-} = +0.36V$.

Temperature = 298 K.

Use 0.059 V as the Nernst factor (2.303RT/F).

The magnitude of change in the cathode potential is x mV and maximum possible van't Hoff factor when precipitate is added to fresh water is y. What is value of (x+y) ?

32. In the presence of excess thiocyanate ion $\left(SCN^-\right)$ the following reaction is first-order in chromium (III) ion, Cr^{3+}; the rate constant is $9.0\times10^{-5}s^{-1}$.

$$Cr^{3+}{}_{(aq)} + SCN^-{}_{(aq)} \rightarrow Cr(SCN)^{2+}{}_{(aq)}$$

If X% completion of reaction is required to obtain a noticeable colour from the formation of the $Cr(SCN)^{2+}$ ion takes in 5 hours what is the value of X ? (ln 5 = 1.62)

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33. The reaction of 4–methyloct-1-ene (P, 2.52 g) with HBr in the presence of $(C_6H_5COO)_2$ gives two isomeric bromides in a 9 : 1 ratio, with combined yield of 50% . Of these, the entire amount of the primary alkyl bromide was reacted with an appropriate amount of diethylamine followed by treatment with $aq.K_2CO_3$ to give a non-ionic product S in 100% yield. The mass (in mg) of S obtained is _____

[Use molar mass (in $g\ mol^{-1}$) : H = 1, C = 12, N = 14, Br = 80]

34. An amount of 3 moles of N_2 and some H_2 is introduced into an evacuated 1 L vessel. The reaction starts at t = 0 and equilibrium is attained at $t = t_1$. The amount of ammonia at $t = 2t_1$ is found to be 34 g. It is observed that $\frac{w(N_2)}{w(H_2)} = \frac{14}{3}$ at $t = \frac{t_1}{4}$ and $t = \frac{t_1}{2}$. What is the value of equilibrium constant(K_C) for decomposition of ammonia represented by chemical equation with stoichiometric coefficient of dinitrogen as 1?

35. How many of the following reactions are endothermic?

(i) $CH_4(g) \rightarrow C(g) + 4H(g)$ (ii) $C_2H_6 + O_2 \rightarrow CO_2(g) + H_2O(l)$

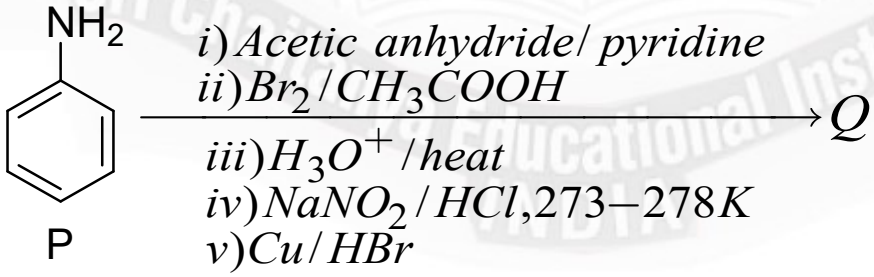
(iii) $Cl(g) + e^- \rightarrow Cl^-(g)$ (iv) $S^-(g) + e^- \rightarrow S^{2-}(g)$

(v) $Ne(g) + e^- \rightarrow Ne^-(g)$ (vi) $Li(g) + e^- \rightarrow Li^-(g)$

(vii) $N^-(g) + e^- \rightarrow N^{2-}(g)$

36. Consider the reaction sequence from P to Q shown below. The overall yield of the major product Q from P is 60%. What amount in grams of Q is obtained from 18.6 ml of P?

(Use density of P = 1.00 $g\,mL^{-1}$, molar mass of C = 12.0, H = 1.0, O = 16.0, N = 14.0, Br = 80.0 $g\,mol^{-1}$)



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MATHEMATICS **Max Marks: 66**

SECTION-I
(INTEGER ANSWER TYPE)

- This section contains SIX (06) questions.
- The answer to each question is a SINGLE DIGIT INTEGER ranging from 0 TO 9, BOTH INCLUSIVE.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual Numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated **according to the following marking scheme:**

Full Marks: +3 If ONLY the correct integer is entered;
Zero Marks: 0 If the question is unanswered;
Negative Marks: -1 In all other cases

37. Let ℓ_1 and ℓ_2 be the lines $\overline{r_1} = \lambda\left(\hat{i}+\hat{j}+\hat{k}\right)$ and $\overline{r_2} = \left(\hat{j}-\hat{k}\right)+\mu\left(\hat{i}+\hat{k}\right)$ respectively. Let X be the set of all the planes H that contain the line ℓ_1. For a plane H, let d(H) denote the smallest possible distance between the points of ℓ_2 and H. Let H_0 be a plane in X for which $d\left(H_0\right)$ is the maximum value of d(H) as H varies over all planes in X. The value of $4\left(d\left(H_0\right)\right)^2$ is _____

38. Suppose that a parabola has vertex $\left(\frac{1}{4},\frac{-9}{8}\right)$ and the equation $y = ax^2+bx+c$, where a > 0 and a + b + c is an integer. The minimum possible value of a can be written in the form $\frac{p}{q},\left(GCD(p,q)=1\right)$ then p + q – 5 is equal to _____

39. The mean and standard deviation of 40 observations are 30 and 5 respectively.
It was noticed that two of the observations 12 and 10 were wrongly recorded.
If σ is the standard deviation of the data after omitting the two wrong observations, then $76\sigma^2-471$ is equal to _____

40. Let $z_1,z_2,z_3,z_4 \in C$ (none of z_i are zero and no two of z_i are equal for i = 1, 2, 3, 4) and C represents set of complex numbers be such that

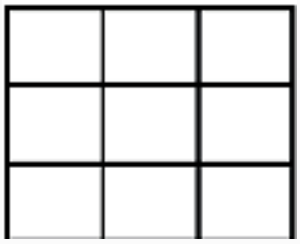
$$z_1\left(z_1-z_2\right)\left(z_1-z_3\right)\left(z_1-z_4\right)=4$$
$$z_2\left(z_2-z_1\right)\left(z_2-z_3\right)\left(z_2-z_4\right)=4$$
$$z_3\left(z_3-z_1\right)\left(z_3-z_2\right)\left(z_3-z_4\right)=4$$

Then the value of $\left|z_1^2+z_2^2+z_3^2+z_4^2\right|$ is equal to _____

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41. In the given figure, each unit square is randomly coloured Red or Green. Let probability that there is no monochromatic (same colour) square of size, 2×2, is p, then value of $\frac{512p}{161}$ is _____



42. Let $f(x)=\begin{cases}0 & if \quad x\le 0\\ P(x) & if \quad 0<x<1\\ 1 & if \quad x\ge 1\end{cases}$ where P(x) is a polynomial of least degree such that f(x) is a differentiable function for all $x\in R$. Area bounded by the graph of y = f(x), with x–axis and between the ordinates x = 0 and x = 1 is A, where 6A is equal to _____

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

•This section contains SIX (06) questions.
•Each question has FOUR options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
•For each question, choose the option(s) corresponding to (all) the correct answer(s).
•Answer to each question will be evaluated **according to the following marking scheme:**
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Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;
Partial Marks: +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: -2 In all other cases.

43. If $A_\alpha=\begin{bmatrix}\cos\alpha & -\sin\alpha\\ \sin\alpha & \cos\alpha\end{bmatrix}$, where $\alpha\in R$; then choose the correct option (s)

A) $(A_\alpha)^{-1}=A_{(-\alpha)}$ **B)** $\det\left(\left(\frac{1}{2}(A_\alpha)\right)^{-1}\right)=2$

C) $A_\pi A_{\frac{3\pi}{2}} A_{\frac{5\pi}{4}} A_{\frac{7\pi}{8}} A_{\frac{9\pi}{16}}\ldots\ldots=I$ **D)** $\text{adj}\left(\text{adj}\left(\text{adj}A_\alpha\right)\right)=A_{(-\alpha)}$

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44. Let $f(x)=\frac{\cos x}{x+\cos x}, x\in(0,\pi)$ then which of the following statement(s) is (are) **CORRECT**?

A) f (x) is monotonically decreasing in $x\in\left(0,\frac{\pi}{2}\right)$

B) f (x) is monotonically increasing in $x\in\left(\frac{\pi}{2},\pi\right)$

C) f (x) has a local maxima at $x=\alpha$ given by $\cot\alpha=-\alpha$

D) f (x) has a local minima at $x=\alpha$ given by $\cot\alpha=-\alpha$

45. Let f be a differentiable function such that f(f(x)) = x, where $x\in[0,1]$. Also f(0) = 1, then value of $\int_0^1 (x-f(x))^{2026}\,dx=\frac{a}{b}$ (where a & b are co-prime), then value of (a + b) is greater than or equal to _____

A) 2029 **B)** 2027 **C)** 2025 **D)** 2026

46. A parabola C with focus (0, 0) and passing through P(3,4). Equation of tangent at P to parabola is 3x + 4y – 25 = 0. A chord through focus S parallel to the tangent at P intersects the parabola at A and B. Which of the following is **CORRECT**?

A) length of AB is 20 units.

B) length of latus rectum of parabola is 20 units.

C) only one real and distinct normal can be drawn from the point $(-3,\ -4)$.

D) only one real and distinct normal can be drawn from the point $(-6,\ -8)$.

47. Let $A={}^{50}C_0-{}^{50}C_3+{}^{50}C_6-----$

$B=-{}^{50}C_1+{}^{50}C_4-{}^{50}C_7------$

The value of A^2+AB+B^2 is m^n, m is prime then which of the following is (are) **INCORRECT**?

A) $\frac{n-1}{m}=16$ **B)** $\frac{n+1}{m}=16$ **C)** $n-m^2=42$ **D)** $n-2m^2=40$

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JEE Advanced 2023 VAVILALA CHIDVILAS REDDY RANK 1 341 360
NEET 2023 BORA VARUN CHAKRAVARTHI RANK 1 720
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48. Consider the function $f(x)=xe^{\frac{-1}{x}}\ (x\neq 0)$, which of the following statement(s) is (are) **TRUE**?

A) $\left(\frac{n}{n+1}\right)^{n(n+1)} < e \quad \forall n \in N$

B) $f(x)f\left(\frac{1}{x}\right) \leq \frac{1}{e^2} \quad \forall\ x>0$

C) Local minimum exists for y = f (x) at x = -1

D) Number of real solutions of $f(x)=f'(x)$ is 1 for $x<0$

SECTION – III
(NUMERICAL VALUE TYPE)

• This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE.**
• For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual Numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
• Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered;
Zero Marks: 0 **In** all other cases..

49. For any real number 'a' let f (a) denotes the maximum value of the function $\left|\sin x+\frac{2}{3+\sin x}+a\right|\ \forall x \in R$. If 'm' be the minimum value of f (a), then '4m' is equal to _____

50. If area bounded by $|x-1|+|y-1|\geq 1$ and $x^2-4x+4\leq 2y-y^2$ is λ, then $\frac{8\lambda}{\pi}$ is equal to _____

51. The number of solutions x of the equation $\sin\left(x+x^2\right)-\sin\left(x^2\right)=\sin x$ in the interval [2,3] is _____

52. A point M moves on the circle $(x-4)^2+(y-8)^2=20$. Then it breaks away from it and starts moving along a tangent to the circle, cutting the x-axis at the point (–2, 0). If $(a,b)\ a,b\in R^+$ be the point on the circle at which the moving point breaks away then $\frac{12a}{b}$ equals

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53. The number of nonempty subsets $S \subseteq \{-11,-10,-9,-8,.....8,9,10,11\}$ that satisfy $|S| + \min(S).\max(S) = 0$ is R, then R is equal to _____

($|S|$ denotes number of elements of set S, min (S) denotes minimum element of set S and max(S) denotes maximum element of set S)

54. Number of ways are there to insert + 's (plus signs) between the digits of 111111111111111 (fifteen 1's) so that the result will be a multiple of 30?

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BELOW 100 ALL INDIA OPEN CATEGORY RANKS	BELOW 500 ALL INDIA OPEN CATEGORY RANKS	BELOW 10 ALL INDIA CATEGORY RANKS COUNT	BELOW 100 ALL INDIA CATEGORY RANKS COUNT	BELOW 1000 ALL INDIA CATEGORY RANKS COUNT
31	95	10	98	579

TOTAL QUALIFIED RANKS FOR JEE ADVANCED-2025 22,094

*DLP/AITS

JEE 2025 STARS SHINE BRIGHT

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ALL INDIA OPEN CATEGORY RANKS



RUTVIK SAI
H.T.No. 256055278 (OBC-NCL)



MAJID MUJAHID HUSAIN
H.T.No. 251134112*



UJJWAL KESARI
H.T.No. 252016104*



AKSHAT KUMAR CHAURASIA
H.T.No. 254065055*

BELOW 100 ALL INDIA OPEN CATEGORY RANKS	BELOW 500 ALL INDIA OPEN CATEGORY RANKS	BELOW 1000 ALL INDIA OPEN CATEGORY RANKS	BELOW 1000 ALL INDIA CATEGORY RANKS COUNT	NUMBER OF QUALIFIED RANKS
29	113	205	745	4,212

*DLP/AITS

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